GAP-10Ω: Reconstruction of the Classical Frame Connection

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Abstract

The covariant-tetrad formulation of UBT uses $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$ and the central anticommutator metric. This note proves two reconstruction results. First, in the nondegenerate metric-compatible torsion-free classical GR branch, the Lorentz frame connection is uniquely the Levi–Civita spin connection of the tetrad. Second, for arbitrary specified torsion, the complete metric-compatible connection is still uniquely reconstructed from the tetrad and torsion through the contorsion tensor. Hence the connection itself carries no residual kinematic ambiguity once (e, T) are fixed. What remains open in full UBT is the *dynamical law selecting the torsion* and the exact representation by which the connection acts on Θ. The flat special-relativistic limit is verified, and the implicit self-consistency equation created by $E_\mu = D_\mu \Theta / \sqrt{\mathcal{N}_0}$ is isolated from the already closed connection-reconstruction problem. The tetrad postulate, Levi–Civita spin connection, and contorsion decomposition used here are standard Cartan/tetrad geometry; the UBT-specific result is their integration into the single-Θ covariant-tetrad architecture and the resulting gap separation.

1 Standard geometric input and UBT-specific scope

The reconstruction results in Sections 2–4 are standard tetrad/Cartan geometry; see, for example, Refs. [1, 2]. They are not claimed as new differential geometry. Their role in UBT is to show that once the anticommutator tetrad and torsion are fixed, Ω_μ is not an additional arbitrary classical field. The UBT-specific open questions are the torsion dynamics, the representation acting on Θ, and the existence of the implicit curved system.

2 Geometric data

Let M be a four-dimensional manifold and let

$$e^a = e_\mu{}^a \, dx^\mu, \qquad \det(e_\mu{}^a) \neq 0, \tag{1}$$

be the real coefficients of the UBT Lorentz-slice tetrad

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k. \tag{2}$$

The induced metric is

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1,1,1,1). \tag{3}$$

Let $e_a{}^\mu$ denote the inverse tetrad.

A Lorentz frame connection is a one-form

$$\omega^a{}_b = \omega_\mu{}^a{}_b \, dx^\mu, \qquad \omega_{\mu ab} = -\omega_{\mu ba}. \tag{4}$$

Its torsion two-form is

$$T^a := de^a + \omega^a{}_b \wedge e^b = \frac{1}{2} T^a{}_{bc} \, e^b \wedge e^c. \tag{5}$$

The corresponding biquaternionic/spin connection is the lift

$$\Omega_\mu = \rho_{\mathrm{spin}*}(\omega_\mu) = \frac{1}{4}\omega_{\mu ab} \, \Sigma^{ab} \tag{6}$$

for the chosen UBT representation. The normalization of the generators Σ^{ab} is representation-conventional; the Lorentz connection $\omega_\mu{}^{ab}$ itself is not.

3 Unique torsion-free compatible connection

Theorem 3.1 (Levi–Civita frame-connection theorem)**.** *For every C^2 nondegenerate tetrad $e_\mu{}^a$, there exists exactly one Lorentz connection $\mathring{\omega}_\mu{}^{ab}$ satisfying*

1. *frame-metric compatibility, $\mathring{\omega}_{\mu ab} = -\mathring{\omega}_{\mu ba}$;*
2. *vanishing torsion, $\mathring{T}^a = 0$.*

It is given by

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right),} \tag{7}$$

where $\mathring{\Gamma}^\rho{}_{\mu\nu}(g)$ is the Levi–Civita connection of Eq. (3).

Proof. The Levi–Civita connection is the unique affine connection satisfying $\mathring{\nabla}_\mu g_{\nu\rho} = 0$ and $\mathring{\Gamma}^\rho{}_{[\mu\nu]} = 0$. Define $\mathring{\omega}$ by Eq. (7). Multiplication by $e_\nu{}^b$ gives the tetrad compatibility identity

$$\partial_\mu e_\nu{}^a - \mathring{\Gamma}^\rho{}_{\mu\nu} e_\rho{}^a + \mathring{\omega}_\mu{}^a{}_b e_\nu{}^b = 0. \tag{8}$$

Antisymmetrising in μ, ν and using $\mathring{\Gamma}^\rho{}_{[\mu\nu]} = 0$ gives $\mathring{T}^a = 0$. Differentiating $g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}$, using $\mathring{\nabla} g = 0$ and Eq. (8), gives $\mathring{\omega}_{\mu ab} + \mathring{\omega}_{\mu ba} = 0$.

For uniqueness, let ω and ω' satisfy both conditions and define $K^a{}_b := \omega^a{}_b - \omega'^a{}_b$. Metric compatibility gives $K_{abc} = -K_{bac}$. Equality of the torsions gives $K^a{}_b \wedge e^b = 0$, hence in the tetrad basis $K_{abc} = K_{acb}$. Combining the two symmetries,

$$K_{abc} = K_{acb} = -K_{cab} = -K_{cba} = K_{bca} = K_{bac} = -K_{abc},$$

so $K_{abc} = 0$ and therefore $\omega = \omega'$. □

4 General metric-compatible connection with torsion

The torsion-free theorem is the special case of a stronger kinematic reconstruction statement.

Theorem 4.1 (Tetrad–torsion reconstruction theorem)**.** *Fix a C^2 nondegenerate tetrad $e_\mu{}^a$ and a torsion tensor $T^a{}_{bc} = -T^a{}_{cb}$. There exists exactly one metric-compatible Lorentz connection with that tetrad and torsion. It is*

$$\boxed{\omega^a{}_b = \mathring{\omega}^a{}_b(e) + K^a{}_b(T),} \tag{9}$$

where $\mathring{\omega}(e)$ is Eq. (7) and, with all frame indices lowered by η_{ab},

$$\boxed{K_{abc} = \frac{1}{2}\left(T_{cab} - T_{abc} - T_{bca}\right).} \tag{10}$$

The convention is $T^a = de^a + \omega^a{}_b \wedge e^b$ as in Eq. (5).

Proof. Write $\omega = \mathring{\omega} + K$. Since $\mathring{\omega}$ is torsion free,

$$T^a = K^a{}_b \wedge e^b. \tag{11}$$

Metric compatibility of both ω and $\mathring{\omega}$ implies $K_{abc} = -K_{bac}$. In frame components the torsion relation is equivalently

$$T_{abc} = K_{acb} - K_{abc} = -(K_{cab} + K_{abc}). \tag{12}$$

Writing the two cyclic permutations of Eq. (12) and taking $T_{cab} - T_{abc} - T_{bca}$ gives $2K_{abc}$, proving Eq. (10). Thus existence is explicit. If two metric-compatible connections have the same tetrad and torsion, their contorsions have zero difference by the same formula, proving uniqueness. □

Corollary 4.2 (No residual kinematic connection ambiguity)**.** *For a nondegenerate tetrad, a metric-compatible connection is completely fixed once the torsion is specified. In particular, in the classical GR branch $T = 0$ implies $K = 0$ and $\omega = \mathring{\omega}(e)$.*

This result does *not* say that UBT has already derived $T = 0$. It says that the remaining full-UBT question is the dynamical law for torsion, not an unconstrained freedom in Ω_μ itself.

5 Gauge covariance and Lorentz-slice preservation

Under a local Lorentz transformation $e \mapsto \Lambda e$,

$$\omega \mapsto \Lambda \omega \Lambda^{-1} - d\Lambda\, \Lambda^{-1}. \tag{13}$$

This is gauge covariance, not a new physical ambiguity. Since $\omega_\mu \in \mathfrak{so}(1,3)$, parallel transport preserves η_{ab}. Therefore the induced vector action preserves the real Lorentz slice and the central anticommutator metric. This closes the connection part of GAP-10L for every metric-compatible branch, including branches with contorsion. It does not prove that the full Θ equations and their sources dynamically keep $D_\mu\Theta$ in that slice.

6 Flat limit

Corollary 6.1 (Special relativity). *For the inertial tetrad $e_\mu{}^a = \delta_\mu{}^a$ in Cartesian Minkowski coordinates,*

$$\mathring{\Gamma}^\rho{}_{\mu\nu} = 0, \qquad \mathring{\omega}_\mu{}^{ab} = 0, \qquad \Omega_\mu = 0, \qquad D_\mu = \partial_\mu, \tag{14}$$

provided the selected UBT vacuum torsion is zero. More generally, flat spacetime may have nonzero connection coefficients in a curvilinear frame, while its curvature remains zero.

7 Implicit Θ–connection self-consistency

The reconstruction theorems treat (e, T) as known geometric data. In UBT, however,

$$E_\mu = \mathcal{N}_0^{-1/2} \left(\partial_\mu \Theta + \rho_*(\Omega_\mu[E, T])\Theta\right) \tag{15}$$

is an implicit nonlinear first-order system. Unique reconstruction of Ω_μ does not by itself prove that Eq. (15) has a local or global solution for every required tetrad.

Moreover, the representation ρ_* matters. A naive one-sided action on an invertible Θ creates a flatness obstruction in a torsion-free compatible branch, whereas a natural two-sided biquaternionic bimodule action can support nonzero conjugate left/right curvatures. This is proved separately in `canonical/gr_closure/gap_10i_integrability_selection.tex`. A subsequent audit proves that the pure Lorentz-compatible representative $A = \Omega$, $B = -\Omega^\ddagger$ has a torsion-free obstruction: with $K = 0$ it implies a concurrent vector and excludes the non-flat Schwarzschild vacuum exterior with nonzero mass. A companion theorem then constructs a local single-Θ representer for every smooth tetrad using explicit composite metric-compatible contortion inside the same pair. See `canonical/gr_closure/gap_10i_paired_con` and `canonical/gr_closure/gap_10i_torsionful_local_representer.tex`.

Defensible status

GAP-10Ω-KIN: CLOSED [L1]. For every nondegenerate tetrad and specified torsion, the metric-compatible frame connection is uniquely $\omega = \mathring{\omega}(e) + K(T)$, up to local Lorentz gauge.

GAP-10Ω-GR: CLOSED [L1]. In the torsion-free classical GR branch, $K = 0$ and the frame connection is uniquely the Levi–Civita spin connection of the tetrad.

GAP-10T-PALATINI: CLOSED CONDITIONALLY [L1]. In the minimal Hilbert–Palatini branch the connection equation is an invertible algebraic Cartan equation. Zero spin current gives zero torsion; specified spin current gives unique contorsion. See `gap_10t_palatini_torsion_dynamics.tex`.

GAP-10T-DYN: NARROWED. The canonical UBT action must derive that minimal branch, its exact paired spin current and normalization, and decide whether extra torsion invariants are present.

GAP-10L-CONN: CLOSED [L1]. Any metric-compatible Lorentz connection preserves η_{ab} and the Lorentz slice. Full dynamical slice preservation remains GAP-10L-DYN.

GAP-10I-PRESCRIBED: CLOSED [L1]. For specified (E_μ, A_μ, B_μ), existence and path independence are exactly controlled by augmented holonomy; see `gap_10i_augmented_holonomy.tex`.

GAP-10I-TORSION-LOCAL: CLOSED LOCALLY [L1]. Composite contortion provides an explicit local curved right inverse of the tetrad map.

GAP-10I-CURVED: LOCAL KINEMATICS CLOSED; DYNAMICS/-GLOBAL PART NARROWED. Canonical on-shell selection, physical torsion constraints, regularity, and global continuation of the implicit system remain to be proved.

References

[1] R. M. Wald, *General Relativity*, University of Chicago Press (1984).

[2] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, "General Relativity with Spin and Torsion: Foundations and Prospects," *Rev. Mod. Phys.* **48**, 393–416 (1976).